

PROJECT OUTLINE

BEng

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**Real-World Autonomous Indoor Delivery Robot with SLAM-
Based Navigation**

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1. INTRODUCTION

1.1 *Project justification*

Robotics has been a large contributor to the rising productivity of warehouses and factories worldwide. Businesses use robots to aid in quality checks and transportation of products from one location to another which reduces human workload, this allows human workers to spend more time on other higher-level tasks instead.

Many of these autonomous robots rely heavily on predefined maps, guiding lines or external markers, these reduces the robot's ability to adapt to environmental changes. A solution to this problem would be to use SLAM as a navigation tool, SLAM enables a robot to build a map while estimating its position within it, which allows for autonomous navigation in unknown indoor areas.

This project proposes the design of a delivery robot that is capable of performing autonomous indoor delivery navigation tasks using a 2D LiDAR sensor and SLAM navigation algorithms. Instead of using existing educational robots such as the TurtleBot3, the proposed robot will be based with a Raspberry Pi 5 single-board computer as it can operate without ROS and also reduces the software complexity.

1.2 *Project Aims and Objectives*

The aim for this project is to design and create an autonomous indoor delivery robot that is capable of mapping, localization and navigation with LiDAR-based SLAM on a Raspberry Pi 5. An example of general use of this is to transport smaller objects, such as small trinkets and water bottles, for convenient collection of refreshments for workers in an office setting.

The robot will be a two-wheeled robot with a leading caster wheel, with a platform or drawer-like container near the top of the robot body for delivery item storage, the robot would also have a LiDAR camera sensor at the top-most part of the robot which is the main contributor for SLAM-based navigation which helps to estimate its location within the office and move from one destination to another.

The robot would have a Raspberry Pi 5 as the base microcomputer which holds the codes and controls for the parts and will connect to other components such as the motor drivers, wheel encoders and the LiDAR sensor.

Below are the objectives for this project:

- Design a 3D printed robot body
- Integrate Raspberry Pi with LiDAR, Motor Driver and Wheel Encoders
- Develop robot software to generate indoor map with SLAM
- Implement autonomous path planning between destinations
- Evaluate system performance such as mapping accuracy and navigation reliability

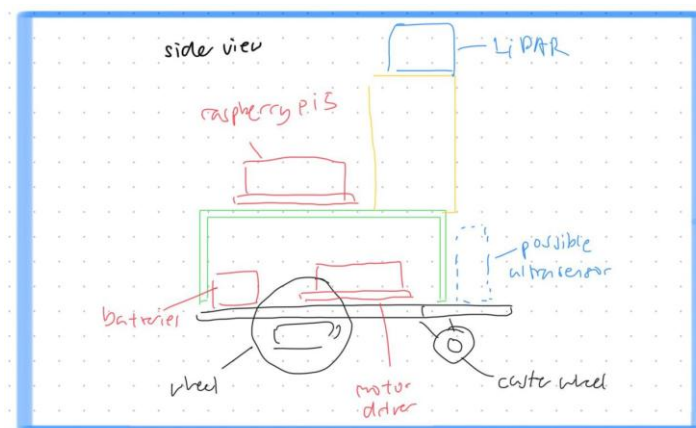
1.3 Methodology

1.3.1 Research

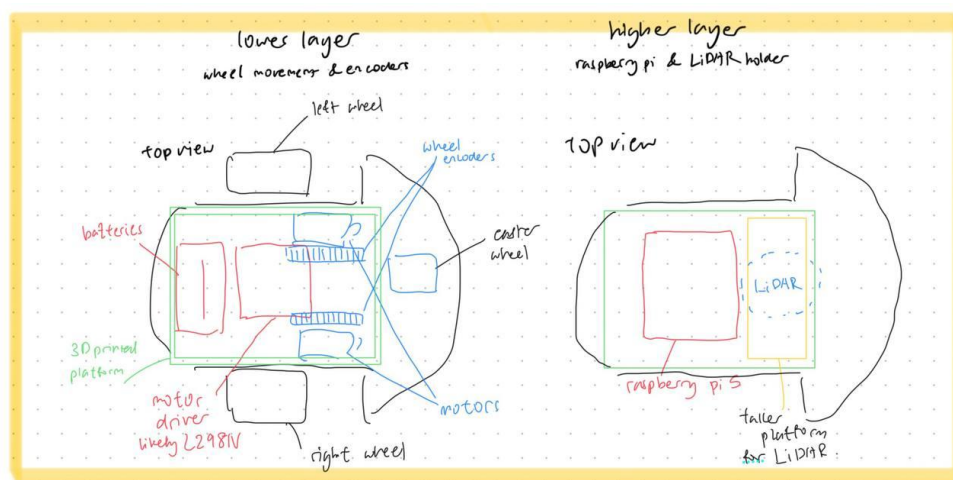
There are some existing similar robot projects such as a Real-World Autonomous Indoor Navigation using SLAM with a TurtleBot3, where they used the Robot Operating System (ROS) framework for TurtleBot3 and had it to navigate an indoor environment with SLAM, using Gazebo to simulate and test the robot code within a controlled online environment before running the code on TurtleBot3. The expected function of this robot is to head to the first set goal, and if it is unable to reach the goal it will return a false signal and move on to the next set goal.

Unlike the referenced project, this proposed system will avoid the use of ROS and instead implement the navigation software directly in Python on the Raspberry Pi platform. The referenced project will therefore be used only as a source of design ideas and hardware architecture.

Below is a proposed sketch that contains a general idea of what the robot may look like, describing the robot from the side-view and top-view in layers:



Side View



Top Views in Layers

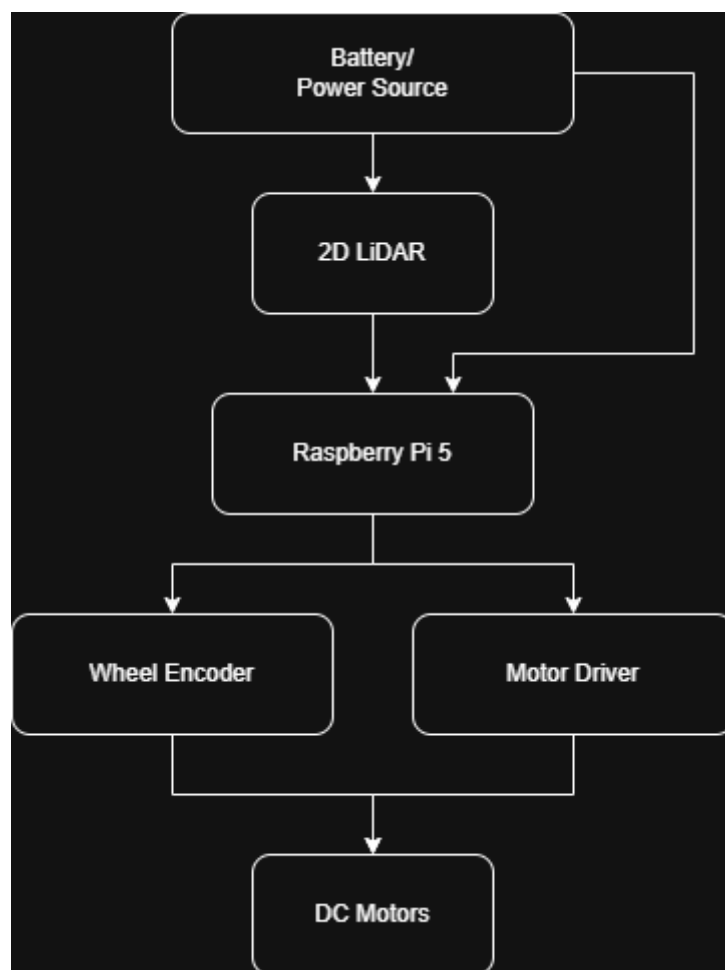
1.3.2 Mechanical and Hardware Development

A 2-wheeled base robot chassis will be loaned; a caster wheel will be attached below to the area that balances the robot, only after adding the 2 larger main motor wheels. The upper robot body will be designed with a 3D modelling software, and many things needs to be taken note of such as the delivery storage area and ensuring the robot body weight is lightweight enough for the robot to move.

The placing of the robot components has to be taken note of too, this robot body has to hold multiple components such as the LiDAR that requires a tall enough height and to also accommodate to the delivery item holder part of the robot, a large enough space for the Raspberry Pi 5 to sit on and to easily access to has to be considered too.

Assembly of the necessary components is required, which are the Raspberry Pi, LiDAR sensor, motors, motor driver, wheel encoders, battery and other electronic components.

Below is a flowchart that describes the general workflow of the robot connections of components.



Hardware Flowchart

The Raspberry Pi 5 will be powered by batteries or a power source wire, connected to it will be the 2D LiDAR sensor camera that sends live distance data to the Raspberry. After the occupancy map and destination is set, the Raspberry will command the Motor Driver to move the DC motors and at the same time it records data from the wheel encoders to ensure the wheels move an estimated distance

towards the destination point.

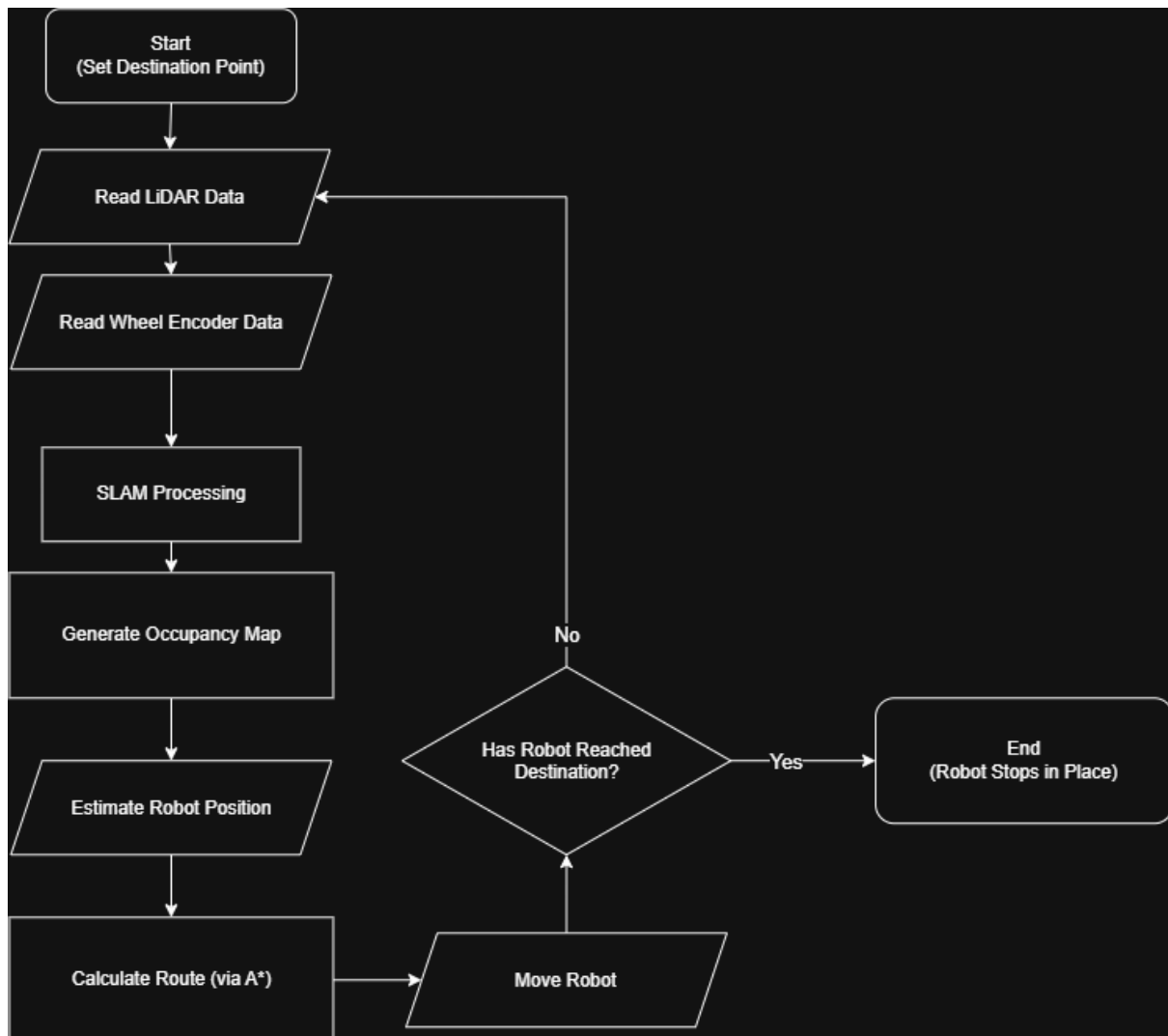
1.3.3 Software Development

Python will be the primary programming language for this robot, to be executed in the Raspberry Pi OS Imager on the Raspberry Pi 5.

Possible data libraries to be used would be BreezySLAM as the SLAM library and RPLiDAR SDK to read the LiDAR data, these would allow the robot to read LiDAR sensor data and generate an occupancy map to estimate its position. There are online tutorials online on how to setup SLAM on the Raspberry Pi 5, the link will be in the References chapter of this report.

The A* path planning algorithm will likely be used for the robot navigation from one point to another, this helps the robot calculate the shortest path from the robot's location to the destination.

Below is a flowchart that describes the general software workflow of the robot.



Software Flowchart

1.3.4 User Interface Development

A graphical user interface will be developed to allow users to control and interact with the robot. The interface will allow users to select a destination, monitor the robot's current position and display the generated occupancy map. It will also provide system status information such as robot movement and delivery completion.

Some software options considered for developing the UI would be PyQt5 or Flask.

1.3.5 Testing and Evaluation

Performance testing will be carried out by assessing mapping accuracy, navigation performance and successful delivery completion. Any software or hardware issues identified during testing will be corrected through iterative improvements. The final system will then be evaluated against the original project objectives.

2. RESOURCE ANALYSIS

2.1 Bill of Materials

Below is a table containing the Bill of Materials required for this project:

Components	Part No.	Quantity	Cost	Description
Raspberry Pi 5	8GB	1	\$266 (Cytron)	Main Microcomputer
MicroSD Storage Card	64GB	1	\$31 (Cytron)	Storage
2D LiDAR Sensor	RPLiDAR A1	1	\$146 (Kuriosity)	360° Scanning
Motor Drivers	L298N	1	\$1.75 (Cytron)	Controls DC Motors
DC Motors		2	\$1.25 (Cytron)	Robot Movement
Wheel Encoders	Rotary Kit	2	\$13.21 (Cytron)	Odometry
Caster Wheel		1	\$2.23 (Amazon)	Robot Balance
Switches	Push Button	~4	\$0.32 (Cytron)	Control Levels
Batteries	3.7V Rechargeable	4	\$25.76 (battery.com)	Portable Power
Resistors	Various Resistances	10	\$0.20 (Cytron)	General Circuit Protection
LEDs	Various Colours	3	\$1.20 (Cytron)	Status Indicators

2.2 Health and Safety

There will not be any use of high-risk equipment for this project, such as any voltages above 30V, high currents at low voltages, very high or low temperatures, hazardous fluids or gases, radiation emitting equipment or being in contact with any hazardous chemicals or substances. The project does contain low-power rotating DC motors, precautions are taken to avoid any accidental injury during testing.

This project also does not involve attaching of any electrodes to any beings, there are no substances

that are hazardous to health, and there are no flammable substances to be used.

This project will not impose any danger to anyone that may cause injury, death or other health-related issues.

2.3 Ethics

This project will not involve any human subjects, and no surveys, interviews or experiments will be done on other people. There will be no collection of data either as no surveys will be conducted.

3. PROJECT PLAN

3.1 Project Work Description

- Order hardware (~ a month before Part B)
 - Order all components online before the start of Part B of the project.
- Download Software needed (a day)
 - Downloading of software required to do the project (Raspberry Pi Imager, BreezySLAM libraries etc.)
- Gather and Assemble hardware (a day)
 - Gather all components and ensure that no needed component is missing, make a list of components used and keep all parts in a dedicated box.
- Motor control (< a week)
 - Make basic motor control system setup and test code with it, goal is to get the motor driver to successfully control both wheel motors.
- Wheel encoder testing (< a week)
 - With previously made system setup for motor control, implement wheel encoders to track the estimate distance the robot had moved.
- LiDAR integration (~ a week)
 - Continuing with the same setup, integrate LiDAR sensor to measure distances around the robot body.
- SLAM implementation (~ a week)
 - Integrate SLAM together with LiDAR measurements.
- Map Generation (~ a week)
 - Continuing on with SLAM implementation with the LiDAR, ensure that a map can be generated and viewed on the UI
- Localization and Navigation (~ a week)
 - Ensure that the robot can localize itself within the occupancy map it generated and navigate to its destination.
- Design 3D Robot Body Chassis (< a week)
 - Design a 3D robot body that will hold the components and also the delivery items.

- 3D Printing the Chassis (~ a week)
- Assemble All Robot Parts and Components (< a week)
 - Assembling the components with the robot base chassis and the 3D printed robot body.
- Testing Assembled Robot (< a week)
 - Ensure that assembled robot works well, fix any minor flaws in chassis.
- Report Documentation (~ a week)
- Prepare Presentation Slides (~ a week)

4. REFERENCES/bibliography

You are required to provide a list of references used to complete the Project Plan, so we can assess your referencing skills at an early stage of your project. You can use either Harvard or Vancouver referencing style.

Latipov Aziz Bobur ugli, Khusanov Saydobid Dilmurod ugli (2025). REAL-WORLD AUTONOMOUS INDOOR NAVIGATION USING SLAM AND MULTI-GOAL EXECUTION WITH TURTLEBOT3. Zenodo. [online] doi: 10.5281/zenodo.17248853

MathWorks (n.d.). What Is SLAM (Simultaneous Localization and Mapping) – MATLAB & Simulink. [online] [www.mathworks.com](https://www.mathworks.com/discovery/slam.html). Available at: <https://www.mathworks.com/discovery/slam.html>

Raspberry Pi (n.d.). Raspberry Pi Documentation - Getting Started. [online] www.raspberrypi.com. Available at: <https://www.raspberrypi.com/documentation/computers/getting-started.html>

simondlevy (2025). GitHub - simondlevy/BreezySLAM: Simple, efficient, open-source package for Simultaneous Localization and Mapping. [online] GitHub. Available at: <https://github.com/simondlevy/BreezySLAM>

AtsushiSakai (2019). AtsushiSakai/PythonRobotics. [online] GitHub. Available at: <https://github.com/AtsushiSakai/PythonRobotics>

Happy DIY Tech (2025). Visual SLAM VSLAM ROS2 Demo: 3D Mapping & Autonomous Navigation using MentorPi M1 Raspberry Pi 5. [online] YouTube. Available at: <https://www.youtube.com/watch?v=FGNCTK-A-TM>

SLAM robot raspberry pi 4 - Raspberry Pi Forums. (2019). [online] [Raspberrypi.com](https://forums.raspberrypi.com). Available at: <https://forums.raspberrypi.com/viewtopic.php?t=246538>

Flask (2010). Welcome to Flask — Flask Documentation (3.0.x). [online] [Palletsprojects.com](https://palletsprojects.com). Available at: <https://flask.palletsprojects.com/en/stable/>

Qt Documentation. (2019). [online] [Doc.qt.io](https://doc.qt.io). Available at: <https://doc.qt.io/>

[illegible]

APPENDIX B: Project Resources Checklist

Please **PRINT** all information **CLEARLY**

Student's Name:	Choo Zhi Yi
UH Student ID Num:	25034158
Project Title:	Real-World Autonomous Indoor Delivery Robot with SLAM-Based Navigation
Supervisor's Name:	Wong Kim Siong
Date:	31 July 2026

Please enter below your **BEST ESTIMATES** for the information requested. You will not be penalised for small errors and later genuine changes - only for not bothering!

				If yes, please provide details
1	Will materials or components need to be ordered?	<u>Yes</u>	No	Raspberry Pi 5, MicroSD storage card, 2D LiDAR sensor, wheel encoders
2	Will you be designing a PCB or drawing which will require manufacture during your project? (Note that this should only be contemplated where absolutely necessary for the success of the project!)	Yes	<u>No</u>	
3	Will PCB or drawing assemblies already designed and manufactured within the School be required? Please specify:	Yes	<u>No</u>	
4	Will a PC be required for work other than for report writing?	Yes	<u>No</u>	
5	If the answer to 4 is Yes, will you require any special hardware installed/attached (e.g., PROM emulation, USB interface). Please specify:	Yes	<u>No</u>	

6	Will you require specific software, other than word processing? (Please specify):	Yes	<u>No</u>	
7	Will your project require technical staff to make items requiring workshop facilities?	Yes	<u>No</u>	
8	If the answer to 7 is Yes, which of the following workshop activities will be required (please circle): Drilling, Sheet metal forming, Lathe work, Milling, Other (Please specify)			
9	Are there other resources required, not covered above? (Please specify):	Yes	<u>No</u>	

APPENDIX C: Project Risk, Ethics and Threat Assessment Checklist

- Please enter below your **BEST ESTIMATES** for the information requested. If yes, please provide details.

1	Is your project likely to involve either deliberate, or possible accidental use or contact with: Voltages above 30V High currents at low voltage (e.g., lead acid batteries) Rotating machines High/Low temperatures Hazardous fluids or gases Radiation emitting equipment Chemicals or Substance Hazardous to Health	Yes	<u>No</u>	
2	Will your project involve attaching electrodes to yourself or someone else in such a way as to produce low impedance contact?	Yes	<u>No</u>	
3	If the answer to 1 and/or 2 is Yes, has an assessment of the risk in the planned procedure been carried out?	Yes	No	
4	If the answer to 3 is Yes, does the analysis show an acceptable level of risk?	Yes	No	
5	Are any substances classified as being 'Hazardous to Health' likely to be used in the course of your work?	Yes	<u>No</u>	
6	Are any flammable substances likely to be used in the course of your work? (e.g., volatile cleaning agents, paint, etc.)	Yes	<u>No</u>	

7	Are there any other aspects of your projected work which might impose a danger to yourself or to others? (Please specify):	Yes	<u>No</u>	
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- Notes: If the answer to any of these questions is YES, you must complete a School of Physics, Engineering and Computer Science risk assessment. This should be approved and signed by your supervisor prior to submission. The risk assessment should also be submitted as an appendix in your report.

Ethical Issues			
8	Will your work involve human subjects, e.g., survey, interview, experiment, etc. This includes friends and family.	Yes	<u>No</u>
	If yes, please provide details		

- If the answers to either of questions 2 or 8 are YES, the proposed work must be submitted for approval to the Ethics Committee before work proceeds. Consult your supervisor and review the [UH Ethics Approval Canvas Module](#) for information on the procedures.
- Please ensure your project **does not** use animals for experimentation or other testing/data gathering. These require a different set of approval procedures which will be addressed on a case-by-case basis. If you believe that you MUST use animals in your project, please contact your Supervisor to discuss this necessity.

APPENDIX D: Industrial Projects (NOT Applicable)

Industrially based project proposals may be submitted by industrialists, part-time students and sandwich students returning from their year in industry. For these projects, an industrial guarantor is required.

Industry based projects are particularly important for part-time students who do not have official timetabled project sessions at the University and are therefore expected to undertake the bulk of their project at their place of work. Part-time students in their second year are encouraged to make early investigations into the possibility of undertaking a project at their place of employment.

Industrial guarantor's agreement:

This form is relevant to those projects that are being undertaken in industry. For an industrial project to run, this form must be completed and signed by the industrial guarantor and the student.

The role of the industrial guarantor:

All final year students are required to undertake an engineering project as part of their degree. Some students, particularly those undertaking a part-time course while working in industry, choose to undertake projects in an industrial environment. For this group of students, each will have an academic Supervisor at the University and an industrial guarantor at the workplace.

The role of the industrial guarantor is to provide the student with technical support in the workplace, and to guarantee that the student has undertaken the work as reported and without undue assistance. The support will nominally require 10-15 hours per week for the duration of the project for part time students.

Part-time students entering their final year who have not arranged a project and guarantor must do so by the start of that academic session, at the latest.

Agreement:

Before completing this section of the agreement, the prospective industrial guarantor should discuss the project with the student and broadly understand the terms of reference of the project and the student's role in the project.

The letter on, company notepaper, should be posted to the School Project Tutor and should contain the wording given below before the project work is started.

I (industrial guarantor's name printed) _____

agree to my role as industrial guarantor for the (student's name printed)

_____.

Signed (guarantor):		Date:	
Contact address & telephone number of guarantor:			
Signed (student):		Date:	
Contact address & telephone number of student:			